

BAKCHAR, Russian Federation

WMO: 293280

Lat: 57.0078N

Lon: 82.0603E

Elev: 110

StdP: 100.01

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -38.6	(c) -35.3	(d) -41.3	(e) 0.1	(f) -38.4	(g) -37.9	(h) 0.1	(i) -35.1	(j) 9.1	(k) -10.1	(l) 8.3	(m) -9.0	(n) 0.9	(o) 10	(p) 0.640

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 10.8	(c) 28.5	(d) 18.6	(e) 26.8	(f) 18.0	(g) 25.2	(h) 17.2	(i) 20.4	(j) 25.7	(k) 19.4	(l) 24.6	(m) 18.4	(n) 23.2	(o) 3.2	(p) 30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.6	13.6	22.8	17.5	12.7	21.8	16.5	11.9	20.8	59.5	25.8	56.2	24.6	52.8	23.1	24.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.4	7.3	6.3	-41.7	31.2	3.3	1.7	-44.0	32.4	-45.9	33.4	-47.8	34.4	-50.1	35.7
			-41.5	21.9	2.9	1.0	-43.6	22.6	-45.3	23.2	-46.9	23.8	-49.0	24.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	0.4	-19.5	-15.5	-6.8	2.1	9.7	15.9	17.8	15.0	8.7	1.7	-8.9	-16.0		
	DBStd	14.33	8.95	7.91	6.69	5.90	5.56	4.28	3.23	3.63	4.40	5.21	8.36	9.05		
	HDD10.0	4188	913	715	521	242	77	7	0	4	76	261	568	806		
	HDD18.3	6603	1172	948	779	486	273	94	49	114	289	517	818	1064		
	CDD10.0	696	0	0	0	6	67	183	242	159	37	3	0	0		
	CDD18.3	70	0	0	0	0	4	21	33	12	1	0	0	0		
	CDH23.3	808	0	0	0	1	88	259	325	124	11	0	0	0		
	CDH26.7	158	0	0	0	0	20	54	63	21	1	0	0	0		
(14) Wind	WSAvg	2.9	3.0	3.2	3.5	3.3	3.2	2.5	2.2	2.3	2.6	3.0	3.2	3.2		
(15) Precipitation	PrecAvg	601	35	25	33	34	47	68	76	69	48	56	61	50		
	PrecMax	800	65	42	75	71	108	116	128	135	89	103	97	97		
	PrecMin	447	4	2	4	10	12	26	13	27	15	26	27	15		
	PrecStd	86	16	9	17	14	23	25	27	23	18	21	18	17		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-0.5	1.2	8.6	20.6	29.3	30.4	30.7	29.1	24.8	17.8	6.5	1.0		
		MCWB	-1.4	0.1	3.1	11.0	15.4	18.5	20.4	19.8	16.3	10.0	5.0	0.2		
	2%	DB	-2.6	-0.8	5.7	17.1	25.7	28.2	28.4	26.6	21.6	14.4	4.4	-0.3		
		MCWB	-3.4	-1.9	1.5	9.3	14.1	18.0	19.5	18.4	14.2	8.8	2.9	-0.8		
	5%	DB	-4.6	-2.4	3.6	14.0	23.0	26.4	26.7	24.4	18.9	11.5	2.4	-2.0		
		MCWB	-5.2	-3.4	0.6	7.7	13.1	17.6	18.6	17.5	13.0	7.5	1.3	-2.7		
	10%	DB	-6.4	-4.2	1.9	11.1	20.0	24.4	25.1	22.4	16.4	8.9	0.8	-4.0		
		MCWB	-7.0	-5.1	-0.2	5.9	11.5	16.9	17.9	16.5	11.5	6.1	0.1	-4.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-1.4	0.2	3.7	12.1	16.9	21.0	22.2	21.1	17.0	11.2	5.3	0.5		
		MCDB	-0.8	1.1	7.8	18.2	25.3	26.8	28.2	26.3	22.2	15.0	6.1	0.8		
	2%	WB	-3.3	-1.8	2.2	9.9	15.3	19.7	20.8	19.6	14.9	9.7	3.1	-0.9		
		MCDB	-2.6	-0.9	4.5	16.5	23.9	25.6	25.8	24.9	20.1	13.3	4.3	-0.4		
	5%	WB	-5.2	-3.3	1.1	8.0	13.9	18.6	19.8	18.3	13.5	7.8	1.4	-2.7		
		MCDB	-4.6	-2.5	3.2	13.7	21.2	24.2	24.7	22.5	18.0	11.3	2.2	-2.1		
	10%	WB	-7.0	-5.2	-0.1	6.1	12.4	17.5	18.9	17.1	12.1	6.2	0.2	-4.6		
		MCDB	-6.4	-4.2	1.9	10.8	19.2	22.8	23.6	21.2	15.6	8.6	0.9	-4.0		
(35) Mean Daily Temperature Range	MDBR		8.9	10.1	10.9	10.4	11.8	11.4	10.8	10.7	9.7	7.2	6.6	7.7		
	5% DB	MCDBR	7.4	6.7	11.0	14.9	16.3	15.3	13.8	14.3	14.3	12.1	5.6	6.8		
		MCWBR	7.3	6.2	7.9	8.4	7.2	6.8	6.1	7.1	7.7	7.3	5.1	6.6		
	5% WB	MCDBR	7.3	6.3	8.7	14.2	14.4	12.9	11.8	12.0	12.7	10.9	5.5	6.6		
		MCWBR	7.2	6.1	6.7	8.4	7.5	6.2	5.8	6.6	7.3	7.2	5.1	6.5		
(40) Clear-Sky Solar Irradiance	taub		0.284	0.318	0.368	0.398	0.402	0.406	0.442	0.408	0.367	0.346	0.312	0.286		
	taud		2.014	1.996	1.933	2.094	2.243	2.308	2.196	2.282	2.351	2.260	2.085	1.962		
	Ebn at Noon		547	692	754	807	834	835	790	794	775	673	508	415		
	Edh at Noon		62	98	136	136	125	119	131	113	91	76	58	50		
(44) All-Sky Solar Radiation	RadAvg		0.74	1.73	3.19	4.44	5.08	5.75	5.21	4.13	2.74	1.37	0.64	0.43		
	RadStd		0.08	0.12	0.22	0.37	0.50	0.60	0.41	0.38	0.36	0.12	0.12	0.07		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page